

# Savir Singh

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## Education

### University of Toronto

HBSc, Computer Science & Mathematics | CS GPA: 3.85 / 4.00

Toronto, ON

2025 – 2029

- Coursework: Real Analysis, Abstract Linear Algebra, Intro to Computer Science, Statistics & Data Science, Reasoning for Computer Science, Microeconomics, Linguistics, Management

## Experience

### KPMG

AI Engineering Intern

Toronto, ON

May 2026 – Present

- Developing LLM-powered agents for automated client query handling while meeting security and compliance requirements
- Supporting policy and platform security teams with audit documentation and regulatory evidence collection
- Refining retrieval-augmented generation (RAG) data workflows to improve database quality and downstream AI reliability
- Redesigned an LLM agent from iterative Q&A to batch spreadsheet processing, covering entire datasets instead of single queries

### General Robotics

Software Engineer Intern, AI Model Research

Kirkland, WA

Sep 2024 – Dec 2024

- Expanded a robotic perception stack with 3 machine learning models for action recognition and visual odometry
- Improved visual odometry performance by 15% through Princeton DPVO pipeline integration
- Built classification and data-quality tooling for robotic sensor datasets and model-selection workflows

### University of Waterloo

Research Assistant & Software Intern

Waterloo, ON

Jan 2024 – Apr 2024

- Supported the FuzzSlice software security research lab under Dr. Nagappan as a high school intern
- Added C++ repository support to FuzzSlice, increasing dataset coverage by 30%
- Built a Flask + React demo platform presented at ICSE 2024 to 2,000+ attendees

## Selected Projects

### Clonio AI

📱 App Store

2024 – Present

Python • Flask • SwiftUI • LLMs • Gemini • NLP • Data Pipelines

- Built an AI personality-cloning platform with a Python Flask backend, iOS and web clients, and custom inference pipelines
- Reached Top 25 in the Productivity category on the App Store in the United States, Canada, and Switzerland
- Scaled the system to 1M+ processed messages
- Drove product traction through 10+ media features and a \$3,500 Ontario government grant for development

### Math Contest Repository

🌐 Live Platform

2023 – Present

Python • Flask • Full Stack • Data Modeling

- Built a large-scale online math contest platform with 500+ users, 6,000+ solution submissions, and 1,400+ problems
- Designed leaderboard systems, automated problem-tagging pipelines, and personalized problem recommendations
- Scaled to 10K+ monthly impressions and partnered with Desmos for free API and premium calculator integration

### Volcano Judge

🔄 GitHub

2022 – 2023

C++ • Python • Sandboxing • Distributed Evaluation

- Developed an online judge for automated code evaluation supporting Python 3 and C++ 20
- Implemented isolated Python sandboxing and REST APIs for scalable grading workflows

### Independent Research

2024 – Present

- Researching odd perfect numbers and independently derived Euler structural characterization
- Collaborated with Carnegie Mellon professor on heuristic algorithms for efficient addition-subtraction chain generation

## Technical Skills

**Languages:** Python, C, C++, Java, SQL, Swift, JavaScript, R, HTML/CSS,  $\text{\LaTeX}$

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, NLP, LLMs, APIs, Statistical Modeling, Reinforcement Learning

**Systems/SWE:** Algorithms & Data Structures, Linux, Git, Docker, Flask, Django, React, ServiceNow, Distributed Systems

## Achievements

Top 5%, Canadian Computing Competition Senior (2024-2025)

\$200 Replit Prize Winner, Hack the North 2023

3x First Place, PEO Programming Competition (2021-2023)

Gold & Silver Medalist, Halton Skills Coding (2023, 2025)